

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to SF or NYC relocation + occasional travel

PROFESSIONAL SUMMARY

Applied AI engineer (May 2025 M.S. Kent State, F-1 OPT) shipping mission-driven LLM applications across healthcare and nonprofit verticals. At Progress Solutions (healthcare consultancy, HIPAA-conscious): agentic LLM workflows with structured reasoning + tool discipline + grounding rules (~25% agent stability gain), RAG with semantic chunking and transformer embeddings (~35% retrieval precision lift, response alignment >90%), and FastAPI/Docker production with structured logging (~30% post-deploy defect reduction). At Suvidha Foundation (nonprofit): transformer-based hierarchical abstractive video summarization across 5,000+ recorded sessions, ~85% alignment with human-curated highlights, 60-70% manual review time reduction, Flask API + lightweight UI for non-technical staff. Founder of E-Farming digital marketplace (PHP/MySQL/Bootstrap, 80-120 active users phase one). Eight shipped/ongoing AI projects including Manga Lens (Chrome Web Store, multi-LLM-provider abstraction with cost-quality trade-off across cloud Claude/GPT and local Ollama, 29 sites) and Agentic Healthcare Claims (5-stage agent pipeline with schema-validated JSON contracts and RAG-grounded CPT/ICD validation). The Beneficial Deployments verticals - healthcare and nonprofit - are the verticals already shipped in.

CORE COMPETENCIES

Mission-Driven Verticals (Healthcare + Nonprofit) • Production Claude API + Multi-Provider LLM Routing • LLM Evaluations + Audit Trails + Grounding Rules • Agent Architecture (Schema-Validated Handoffs) • Cost-Quality Trade-Off (Cloud vs Local Inference) • FastAPI + Docker + Structured Logging • Pair Programming + Non-Technical Stakeholder Comms • Founder Track (E-Farming, Manga Lens Solo Ship) • Teaching + Mentoring (150+ Students at Kent State) • F-1 OPT, Visa-Sponsorship Pathway

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI / ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; agent stability up ~25%; hallucination reduction >30%.
- Built production **RAG systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking and transformer-based embeddings improved **contextual retrieval precision ~35%** and reduced irrelevant retrieval >30%; retrieval-grounded response alignment exceeded **90%**.
- Designed and shipped a **five-agent Healthcare Claims pipeline** (Intake, Validation, Consistency, Duplicate Detection, Risk Scoring) with **schema-validated JSON contracts (pydantic)** between agents to prevent cascading hallucinations - audit-ready reasoning traces for compliance review.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; recall up **15-20%** on high-risk categories with precision held >90% via class weighting, stratified sampling, and threshold calibration.
- Packaged ML / LLM inference as **FastAPI / Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; cut post-deployment defects ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, and support ticket logs; raised dataset reliability above **98%** and cut downstream model instability ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data-lineage documentation, evaluation audit trails, and stakeholder system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering Synthesis Order-to-Cash; reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored-procedure deploys; cut deployment errors >**30%** and improved release-cycle time ~**35-40%**.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization** system over 5,000+ recorded sessions with hierarchical abstractive summarization and timestamp-anchored clip extraction. Manual review time cut **60-70%**; ~**85%** alignment with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinations ~**30%** and lifted contextual answer accuracy >**85%**.
- Deployed the stack as a Flask-based API with a lightweight web interface for non-technical nonprofit staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents preventing cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents. ANN similarity for duplicate detection. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector retrieval (ANN), pgvector, GPT-4 / Claude, schema contracts (pydantic)

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped solo, Chrome Web Store)

Real-time AI manga / webtoon translation extension shipped solo to the Chrome Web Store. Multi-provider abstraction across 4 AI vision providers (Claude Sonnet / GPT-4o mini / GPT-4.1 Nano / Ollama) with per-provider payload routing (WebP cloud, JPEG Ollama after CUDA crashes), 7-day translation cache, 29 site selectors via Manifest V3 service workers, narrowed host permissions, privacy policy.

TypeScript, Manifest V3, Service Workers, Claude / GPT-4o mini / GPT-4.1 Nano / Ollama, npm

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate structured-prompt transformation layer lifted contextual alignment ~30% and first-pass image success ~25-30% over naive direct-prompt orchestration.

FastAPI, React + TypeScript + Vite, Tailwind, Perplexity Sonar, Replicate, Docker

Agentic Pixel Character Synthesis & Animation Engine (ongoing)

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an LLM-based agent orchestrator that decomposes high-level prompts into generation tasks.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, Stable Diffusion, LoRA, ControlNet

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced a two-stage CNN pipeline with a unified YOLOv8 detect-and-classify model; reduced inference latency ~30%. Sliding-window confidence aggregation cut blink-driven false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time operation.

YOLOv8, OpenCV, PyTorch, LabelImg

EDUCATION

Master of Science, Computer Science - <i>Kent State University</i>	<i>Aug 2023 - May 2025</i>
Focus: Applied Machine Learning, NLP, Database Systems.	
Bachelor of Technology, Computer Science - <i>KL University</i>	<i>Jun 2019 - May 2023</i>
Founder / Lead of E-Farming platform (university project, 80-120 active users phase one).	

CERTIFICATIONS

Deep Learning Specialization - <i>DeepLearning.AI (Coursera)</i>	2024
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SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP
AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, Hugging Face Transformers, Diffusers, YOLOv8, OpenCV, ControlNet, Stable Diffusion
LLM / GenAI: RAG, agentic workflows, schema-validated structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding rules, embeddings, vector search (FAISS / pgvector / Pinecone-portable), audit-trace reasoning, multi-agent orchestration, multi-provider AI routing (Claude / GPT-4o / Perplexity / Replicate / Ollama)
Backend: FastAPI, Flask, Node.js (Manifest V3 service workers), REST, Docker, structured logging, load simulation
Frontend: React, Vite, Tailwind, Chrome Extensions (Manifest V3 + content scripts + service workers)
Data: PostgreSQL + pgvector, Pandas, NumPy, ANN similarity, embeddings, SQL data pipelines, T-SQL stored procedures
DevOps: Docker, Jenkins, GitHub Actions, Grafana, AWS/GCP/Azure-portable cloud-ready deployment
Domains: Applied AI, Healthcare AI (HIPAA-conscious), Regulated-Domain LLM Systems, Document Processing + RAG, Multi-Agent Validation, Forward Deploy / Customer-Facing Delivery, Nonprofit ML